



Guidelines

Heparin-Induced Thrombocytopenia (HIT)

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Hospital Wide	Medical, Nursing, Pharmacy

Introduction

This guideline provides guidance to all Sir Charles Gairdner Osborne Park Hospital Care Group (SCGOPHCG) staff involved in the management of Heparin-Induced Thrombocytopenia. This document supports and complies with related legislation, policy and other related documents [here](#).

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☐	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGOPHCG Mission & Values	☐	Staff Safety	☐
Other:	☐		

1.0 General Information

This guideline is to be used in conjunction with advice provided by the haematology team. Call the Haemostasis thrombosis registrar

Heparin-induced thrombocytopenia (HIT) is an adverse drug reaction to heparin that leads to formation of platelet-activating IgG antibodies that target complexes of platelet factor 4 (PF4) and heparin^{1,2,3}. HIT increases morbidity and mortality due to significantly increased risk of thromboembolism (arterial, venous or microvascular e.g. pulmonary embolism, ischaemic limb necrosis, stroke etc.)^{1,2,3}.

2.0 Risk Factors^{5,6}

- Clinical setting: Surgical > Medical setting, rare in obstetric and paediatric patients.
- Heparin type: Unfractionated heparin > low molecular weight heparin (LMWH) (3% vs <1%).
- Female > Male.
- There is a reduced incidence with shorter duration of exposure to heparin (<5 days).

3.0 Diagnosis

HIT is a clinicopathologic diagnosis that requires integration of clinical and laboratory testing. There are several clinical features that are suggestive of HIT. These include:

- Thrombosis (arterial or more commonly venous) – occurring generally 5 days after heparin exposure and up to 30 days after heparin cessation.
- Fall in platelet count 5-10 days after heparin commencement, may occur sooner if recent exposure to heparin (in the last 3 months).
- Fall in platelet count by at least 30-50% from baseline, usually $> 20 \times 10^9/L$.

Less common manifestations include:

- Skin necrosis at the heparin injection site.
- Anaphylactoid reaction to iv heparin bolus
- DIC
- Adrenal haemorrhage secondary to adrenal vein thrombosis
- Limb gangrene

3.1 Clinical probability scoring

The 4Ts score is used to estimate the likelihood (pre-test probability) of HIT. It is used to make a presumptive diagnosis of HIT until laboratory results are available. See table 1⁷The 4Ts has a high negative predictive value (>99%).

Table 1: The 4Ts score, a clinical probability scoring system³

4Ts Category	2 points	1 point	0 point
Thrombocytopenia	Platelet count fall $>50\%$ and platelet nadir $\geq 20 \times 10^9/L$	Platelet count fall 30–50% or platelet nadir 10–19 $\times 10^9/L$	Platelet count fall $<30\%$ or platelet nadir $<10 \times 10^9/L$
Timing of platelet count fall	Clear onset between days 5–10 or platelet fall ≤ 1 day (prior heparin exposure within 30 days)	Fall in days 5–10, but not clear (e.g. missing platelet counts); onset after day 10; or fall ≤ 1 day (prior heparin exposure 30–100 days ago)	Platelet count fall ≤ 4 days without recent exposure
Thrombosis or other sequelae	New thrombosis (confirmed); skin necrosis [§] ; acute systemic reaction post-intravenous unfractionated heparin (UFH) bolus	Progressive or recurrent thrombosis [*] ; Non-necrotising (erythematous) skin lesions [§] ; Suspected thrombosis (not proven)	None
Other causes for thrombocytopenia	None apparent	Possible	Definite
Pre-test probability for HIT:			
Low probability: score ≤ 3 ; Intermediate probability: score 4–5; High probability: score 6–8			
[§] <i>Skin lesions at heparin injection sites.</i>			

Table adapted from: Joseph J, Rabbolini D, Enjeti A, Favaloro E, Kopp MC, McRae S, et al. Diagnosis and management of heparin-induced thrombocytopenia: a consensus statement from the Thrombosis and Haemostasis Society of Australia and New Zealand HIT Writing Group. Med J Aust. 2019;210.

3.2 Laboratory Testing

Once the patient is clinically suspected of having HIT and if the 4T score is >4 , proceed with laboratory testing. This is divided into 2 steps⁶:

1. An immunoassay to detect the presence of the anti-PF4/heparin antibodies.
 - These tests have a high sensitivity (~97%) but limited specificity (~82%).
 - The Stic assay is rapid with results available within an hour.
2. Functional assays to determine if the anti-PF4 /heparin antibodies are pathogenic.
 - These tests have a high sensitivity (95%) and specificity (95%).

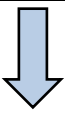
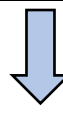
Notes for sample collection:

- For more specialised assays, blood samples collected ≥ 4 hours after the last administration of heparin are preferred.
- Provide one heparin ampoule (with same batch number) for functional assay testing.

- For entire panel of testing, provide fresh samples in **1 Serum (Red) tube and 3 citrate (Blue) tubes**.

See table 2 below for further details on the testing.

Table 2: Characteristics of different immunological and functional assays for HIT^{3,8}

Immunoassays – detects heparin-PF4 antibodies					
Assay	Result	Advantages	Disadvantages	Sensitivity*	Specificity*
STic (Lateral flow assay)	Qualitative	Rapid turnaround time; Highly sensitive; Therefore, good screening test.	Low positive predictive value (<50%); Result by visual inspection i.e. poor inter-observer variability for weak antibodies.	93-100%	51-93%
ELISA [Asserachrom HPIA – IgG (Stago)]	Quantitative	High negative predictive value. Reporting via optical density and cut-off. Possible confirmatory step with high heparin concentration.	Longer turnaround time due to batch testing; Not available for single patient testing due to technical expertise and time required.	100%	65%
 <i>A positive immunoassay should be followed by a functional assay to confirm</i> 					
Functional Assays – measures heparin dependent platelet activation/aggregation					
Assay	Method	Advantages	Disadvantages	Sensitivity*	Specificity*
Multiplate (Heparin-induced multiple electrode aggregometry)	Whole blood. Impedance aggregation	Available on demand if there is a suitable platelet donor. Quick turnaround time.	Requires fresh responsive platelets	84%	95%
Light transmission Aggregometry (LTA)	Platelet rich plasma	Available on demand if there is a suitable platelet donor. Quick turnaround time.	Requires fresh responsive platelets	60%	96%

Serotonin Release Assay is considered the gold standard - not available in WA.

*All sensitivity and specificity data based on Favaloro *et al*⁸, except STic where product insert information is also listed.

*Contact Haematology for advice regarding interpretation of **a positive test**.

4.0 Management of HIT

- STOP ALL HEPARIN** – unfractionated or low molecular weight (e.g. heparin, enoxaparin, dalteparin, nadroparin)³.
 - This includes all devices containing heparin (e.g. syringes, bonded catheters), heparin flushes or line locks and heparin containing products (e.g. some blood derived products such as prothrombinex).
 - Ensure all therapies and devices are checked for heparin content before being administered to the patient.
- Commence alternative therapeutic anticoagulant.**
- HIT is considered a severe/life threatening adverse drug reaction (ADR):
 - Once the diagnosis is confirmed, a clinical alert must be placed on medical records by the treating team to avoid inadvertent exposure to heparin in the

future. To apply /initiate a clinical alert follow the procedure on Healthpoint (see flow chart [here](#) and relevant notification form: [SCGH](#) and [OPH](#)). When completing forms select alert code “D. Other3”, Serious Drug Reactions.

- ALL adverse reactions must be documented:
 - On the WA Hospital Medication Chart (WA HMC); and
 - By providing a red arm band; and
 - In the discharge summary; and
 - Provide an [ADR information consumer leaflet](#) to the patient (for new reactions during admission).
- A medical alert bracelet is also recommended.
- The median time to platelet count recovery is 3-4 days. 90% of patients with HIT have platelet recovery within 7 days. If not, see ‘refractory HIT’ below¹⁹.

4.1 Alternative anticoagulants for Treatment of HIT³

- Treatment is required due to the hypercoagulable nature of HIT.
- Choices include thrombin inhibitors or non-heparin based anti-Xa inhibitors (see [Appendix 2](#)).³
- Prescribe **therapeutic doses** rather than prophylactic doses.¹

Anticoagulant choice is dependent on clinical situation (e.g. renal impairment, need for procedural intervention).

- See [Appendix 2](#) for summary of anticoagulants used to treat HIT.

Anticoagulation selection in Acute HIT

Dependent on 4 clinical criteria

		1) Clinically unstable	2) Life- or limb-threatening ischemia	3) Renal dysfunction CrCl <30ml/min	4) Liver dysfunction Bili >1.5mg/dL
Oral	Oral Xa inhibitors*	●	●	●	●
	Vitamin K antagonist (VKA)	●	●	●	●
IV	Argatroban	●	●	●	●
	Bivalirudin	●	●	● [^]	●
SQ	Danaparoid	●	●	●	●
	Fondaparinux*	●	●	● [~]	●

● Agent is preferred

● Agent is not preferred, but can be considered based on availability, risk/benefit

● Agent is not recommended

ⓘ Laboratory monitoring required

* Existing data with rivaroxaban, apixaban

[^] If argatroban not available, can use with close monitoring due to accumulation risk

^{*} Trivial risk of reported HIT,⁵⁴ but has been demonstrated to be safe for use in acute HIT^{55,56}

[~] Use in renal dysfunction has been reported^{57,58}

Bili, bilirubin; CrCl, creatinine clearance; IV, intravenous; SQ, subcutaneous

Adapted from May et al.⁶

- Refer to individual medication guidelines in the Appendices for more information (e.g. dosing, administration, and monitoring):
 - Danaparoid – monitored IV infusion initially. Can be used in unstable patients and those with renal impairment (with dose adjustment).
 - Fondaparinux – fixed subcutaneous injection dose.
 - Bivalirudin - monitored IV infusion.
 - Direct Oral Anticoagulants (DOACs)- not recommended in unstable patients, or those with life or limb-threatening ischaemia.
 - Argatroban – monitored IV infusion. Requires an IPA. It is not stocked and is unregistered.
 - See special patient populations for managing patients on renal replacement therapy with HIT and managing patients with Acute Coronary Syndrome (ACS) and HIT.

4.1 Transitioning to oral anticoagulation

- Warfarin and other Vitamin K antagonists are **contraindicated in the initial management** of HIT-associated thrombosis due to the risk of venous limb gangrene.³
 - Warfarin should not be started until the platelet count is $\geq 150 \times 10^9/L$. (See 5.1)
 - Patients should continue a non-heparin anticoagulant until INR has reached the therapeutic range.
 - If warfarin has already been started when HIT is diagnosed and the INR is not therapeutic, reverse with intravenous vitamin K (discuss with Haematology first).
- Transitioning to a DOAC as an alternative to warfarin is an alternative option⁶.
 - NOTE: if the patient does not have a PBS indication for a DOAC, an IPA may need to be submitted.
 - HIT with thrombosis: commence loading dose followed by maintenance dose DOAC.
 - HIT without thrombosis: commence maintenance dose (i.e.. Rivaroxaban 20mg daily or apixaban 5mg bd).

4.2 Duration of HIT treatment

- HIT with thrombosis: therapeutic anticoagulation for at least 3 months^{2,3}.
- Isolated HIT without thrombosis: therapeutic anticoagulation until platelet count $\geq 150 \times 10^9/L$ and for a minimum of 4 weeks.^{1, 2,3}
- Refer to Haematology for follow-up once discharged.

4.3 Other precautions and important information

- *Platelet transfusion is not recommended* in HIT in the absence of clinically significant active bleeding³, as there is a concern that the platelets could enhance and increase the thrombotic risk.
- See [Appendix 2](#) and individual medication guidelines for more information on precautions and contraindications for specific anticoagulants.

4.4 Transition to Warfarin^{2,3,10, 11,12}

- If a patient must transition to warfarin, only consider transitioning when:
 - therapeutic parental anticoagulation is stable; **and**
 - the platelet count has risen to $>150 \times 10^9/L$ and is stable.
- Initial warfarin doses should not exceed 5mg¹¹.
 - Avoid larger loading doses to minimise the transient hypercoagulable state induced by the rapid decline in protein C.
- After starting warfarin, parenteral anticoagulant therapy must be continued until the international normalised ratio (INR) is therapeutic and for a minimum of 5 days².

5.0 Management of HIT in Special Patient Populations

5.1 Impaired renal function $<30\text{mL/min}$ and dialysis³

- Most HIT treatment options are relatively contraindicated. Specialist advice required.
- Regional (circuit) anticoagulation
 - alternative options for patients with prior history of HIT include regional citrate, saline flushes, non-heparin anticoagulants and epoprostenol (continuous renal replacement therapy). Specialist advice required.

5.2 Concurrent Acute Coronary Syndrome (ACS)

- Following or during an episode of HIT.

- Refer to Cardiology and the bivalirudin guideline for more information:
 - STEMI undergoing Percutaneous coronary intervention.
 - NSTEMIs +/- subsequent PCI.

5.3 Cardiopulmonary bypass on or off pump³

- Patients undergoing cardiac surgery, vascular surgery, or surgeries that require cardiopulmonary bypass (CPB) following or during an episode of HIT need intraoperative anticoagulation. Discuss options with Haematology.

5.4 Refractory HIT^{3,20}

- Otherwise known as 'persisting HIT'. It refers to persisting thrombocytopenia despite stopping heparin.
- Consider switching to another non-heparin anticoagulant.
- Consider intravenous immunoglobulins (IVIg) use.

5.5 Pregnancy¹²

- HIT is rare in pregnancy.
- Recommended agents are danaparoid (most evidence) and fondaparinux (both Category C).

5.6 Autoimmune HIT^{6,19}

- Autoimmune HIT refers to any HIT-related antibodies that activate the platelets (and cause thrombocytopenia) in the absence of heparin.
- There are several different syndromes associated with aHIT, including delayed onset HIT, persisting HIT, severe HIT with overt DIC, flush heparin HIT and spontaneous HIT.
- The management of aHIT syndromes is the same as classical HIT (ie avoid heparin and use a non-heparin anticoagulant).
- IVIG can be considered in severe cases. Discuss with haematology.

5.7 Prior history of HIT³

- **Generally, avoid heparin for future procedures, VTE treatment or VTE prophylaxis.**
- Transient pathogenic antibodies with median time of disappearance: 50-80 days. Therefore, may not be detectable on subsequent testing.
- Short term heparin can be considered for certain high-risk scenarios (e.g. percutaneous intervention and cardiopulmonary bypass). Always discuss with Haematology as there is a risk of HIT with re-exposure. An alternative agent is bivalirudin.

Related Legislation and Policy Documents

Legislation:

- [Medicines and Poisons Act 2014](#)
- [Medicines and Poisons Regulations 2016](#)
- [Health Services Act 2016](#)
- [Mandatory Policy MP0077/18 Statewide Medicines Formulary Policy](#)
- [SCGOPHCG Formulary Management Procedures MMP013](#)
- [SCGH Hospital Policy HP261 High Risk Medications](#)
- [SCGH Hospital Policy 141 Medicines Management](#)
- [SCGH Medicines Management Procedures and Standards:](#)
- [Nursing Practice Guideline No.51 Medication Management](#)

- [Nursing Practice Guideline No.4 Intravenous Therapy](#)
- NSQHS Standard 1.1.1: An organisation-wide management system is in place for the development, implementation and regular review of policies, procedures and/or protocols.
- NSQHS Standard 1.7: Developing and/or applying clinical guidelines or pathways that supported by best available evidence (all subsequent actions of this criterion are applicable)
- NSQHS Standard 4.1: Developing and implementing governance arrangements and organisational policies, procedures and/or protocols for medication safety, which are consistent with national and jurisdictional legislative requirements, policies and guidelines.
- NSQHS Standard 4.1.2: Policies, procedures and/or protocols are in place that are consistent with legislative, national, jurisdictional and professional guidelines.

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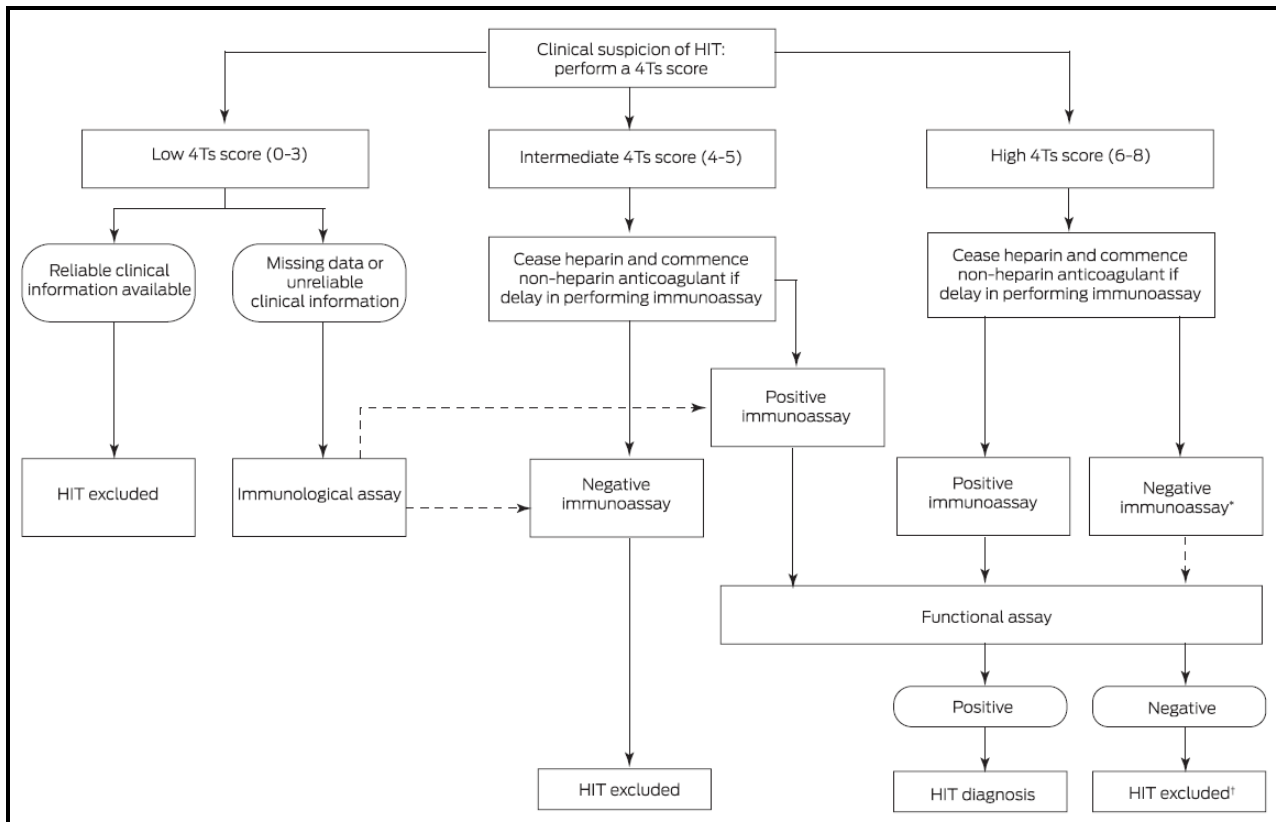
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For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

This document can be made available in alternative formats on request for a person with a disability.

Appendix 1: Suggested diagnostic algorithm for HIT³



*If the immunoassay is negative, it is recommended to treat the patient as not having HIT until further clarification is attained, which can occur with clinical reassessment and/or a functional assay, or if a functional assay is not available, repeating the immunoassay or performing an alternative high sensitivity immunoassay. Seroconversion may take 4-6 days to develop.

† If the ELISA immunoassay is positive and the functional assay is negative, we recommend examining the optical density of the immunoassay result and the sensitivity of the functional assay to guide patient management. Retesting can also be considered. If the clinical suspicion of HIT remains high despite negative testing and/or the patient's clinical condition changes, there is a need to reassess the possibility of HIT and retesting should be considered.

Algorithm image taken from: Joseph J, et al. Diagnosis and management of heparin - induced thrombocytopenia. Med J Aust. 2019;210³.

Appendix 2: Characteristics of anticoagulants used to treat HIT and in patients with prior history of HIT.^{3,13}

	Danaparoid	Fondaparinux	Bivalirudin*	DOACs*	Argatroban (SAS)
Indications per Statewide Medicines Formulary (SMF)*	1 st line for treatment of HIT.	2 nd line for treatment of HIT. 1 st line for treatment of VTE in patients with history of prior HIT. 1 st line for VTE prophylaxis in patients with history of prior HIT.	Current or previous HIT requiring PCI (including NSTEMI with PCI) in combination with aspirin. An IPA is required for all other indications	An IPA is required for indications not meeting the SMF criteria . Option for clinically stable patients at average risk of bleeding. <u>Pros</u> : Oral, effective, convenient. <u>Cons</u> : Minimal evidence currently in limb or life-threatening HIT. Avoid in these situations. Consider drug interactions. ¹	An IPA is required for use – not on formulary and non-stocked medicine. Non-TGA registered and requires SAS application.
Mechanism of action	Mostly anti-Xa	Anti-Xa	Direct thrombin inhibitor (DTI)	Direct Xa inhibitor: Apixaban, rivaroxaban. DTI: Dabigatran	Direct thrombin inhibitor
Half-life	25 hours	17-20 hours	25 minutes	Variable per medicine.	39 – 51 minutes
Therapeutic Monitoring	Anti-Xa	No monitoring/Fondaparinux anti-Xa Please discuss with haematology prior to ordering	APTT , ACT	No monitoring/ Drug specific levels. Please discuss with haematology prior to ordering	APTT
Renal impairment CrCl <30mL/min and renal replacement therapy	Relatively CI. Specialist advice required: Renally cleared, long half-life, not dialysed/or unknown dialysability and no antidote. Requires careful dose adjustments and appropriate monitoring.	Relatively CI. Specialist advice required: Renally cleared, long half-life. Requires careful dose adjustments and appropriate monitoring.	Relatively CI. Specialist advice required if renal function <50 mL/min.	All relatively CI. Specialist advice required	No dose adjustment required.
Hepatic impairment	Relatively contraindicated in severe hepatic insufficiency. Specialist advice required	No dose adjustment required	No dose adjustment required	All relatively CI. Specialist advice required.	Reduce dose in moderate to severe hepatic impairment (Child Pugh B and C)
Clearance	Renal	Renal	Renal and proteolytic cleavage	Refer to individual monographs on MIMS	Hepatic
Impact on PT/INR	Minimal	Not affected	Yes	Refer to individual monographs on MIMS	Yes

CrCl = creatinine clearance, DOACs = direct acting oral anticoagulants. CI=contra-indicated.

**Bivalirudin and DOACs are not listed on the WA Statewide Medicines Formulary for treatment of HIT at time of endorsement. Check medicine restriction as Individual Patient Approval (IPA) may be required.

Appendix 3: Danaparoid^{1,3,13}

1.0 Availability

- 750 anti-Xa units/0.6mL danaparoid sodium, ampoules. (Orgaran®).

2.0 Indications

- On the SMF as 1st line for treatment of HIT. Consult Haematology.

3.0 Dose and Administration

3.1 Intravenous

- Bolus dose: Dilute bolus dose with 0.9% sodium chloride or 5% glucose if needed.

Weight	IV Bolus Dose
< 60 kg or moderate to severe renal impairment	1500 units
60 – 75 kg	2250 units
75 – 90 kg	3000 units
> 90 kg	3750 units

- Seek haematology advice if there is concern with administering a bolus.

Intravenous maintenance infusion:

- Prepare infusion: dilute 3750 units (3 mL) to 50 mL with 0.9% sodium chloride or 5% glucose in a syringe to give 75 unit/mL. Infuse via a syringe pump.
- Infusion Rate: 400 units/hour for 4 hours then, 300 units/hour for 4 hours then, 200 units/hr (150 units/hr in renal insufficiency). Maintenance infusion (range 100 to 400 units/hour) adjusted according to anti-Xa level.

Monitoring

Target anti-Xa level: 0.5 to 0.8 units/mL.¹ (Request form should be marked patient receiving danaparoid infusion).

- Anti-Xa levels should be checked 8 hours after initiation and 4 hours after any adjustment or disruption to therapy.
- Once dose is stable, anti-Xa levels must be measured daily, as a minimum.

Anti-Xa level	Action	Repeat Anti-Xa level
> 1.0 units/mL	Reduce dose by 50%	4 hours later
< 0.5 units/mL	Increase dose by 20%	4 hours later

3.2 Subcutaneous

Danaparoid is 100% bioavailable via subcutaneous administration and this can be considered once the patient is stable, on the advice of the Haematologist.

- Calculate the total 24-hour IV danaparoid requirement (based on current stable infusion rate), divide by 2 and give this dose subcutaneously TWICE daily. (e.g. if the patient is running danaparoid IV infusion at 200units/hr and anti-Xa levels stable, calculated 24-hour dose is 200 units x 24 hrs = 4800 units, divided by 2 gives the subcutaneous dose of 2400 units 12 hourly.)

Subcutaneous Danaparoid Monitoring

- Target anti-Xa levels 0.5 to 0.8 units/mL at 6 hours post dose. (Request form should be marked patient on subcutaneous danaparoid).
- Contact Haematology for advice on dose adjustments.

4.0 Renal impairment < 30 mL/min and renal replacement therapy

- Use is relatively CI. Danaparoid is renally cleared, has a long half-life, is not dialysed /or has unknown dialysability, and there is no antidote. Requires dose adjustments and careful monitoring. Contact Haematology for advice on dose adjustments and monitoring.

Appendix 4: Fondaparinux^{3,13,14,15,16}

1.0 Availability

- 2.5mg fondaparinux in 0.5mL pre-filled syringe (Arixtra®).

2.0 Indications

- On the SMF for 2nd line treatment of HIT, and
- For patients with a prior history of HIT.

3.0 Dose for Treatment of HIT

Weight	Once Daily Dose (subcutaneously)
< 50 kg	5 mg
50-100 kg	7.5 mg
> 100 kg	10 mg

4.0 Therapeutic Drug Monitoring

- Generally, not required.
- There is no validated Anti-Xa therapeutic range for fondaparinux. Therefore, fondaparinux levels may only be taken and requested if approved by Haematology.
- If monitoring is approved, request for a “fondaparinux-specific-anti-Xa level”
 - Reported mean therapeutic dosing fondaparinux-specific-anti-Xa level 3 hours post dose in product information is 1.2-1.26 mg/L.

5.0 Renal Impairment <50 mL/min and Renal Replacement Therapy

- Fondaparinux is mainly excreted by the kidney (~77%). It is relatively contraindicated in CrCl <30 mL/min and must be used with caution in CrCl 30 – 50 mL/min due to risk of accumulation and bleeding.
- Reassess renal function regularly.
- Clearance of fondaparinux increases by up to 20% during haemodialysis.
- For renally compromised patients who require therapeutic dosing contact Haematology for advice on dose adjustments and monitoring.

Creatinine Clearance (CrCl)	Recommendations for dose adjustment – For Haematology use only
30-50 mL/min	Total clearance is reduced ~40% compared to patients with CrCl >80 mL/minute. ¹⁴ Use with caution with dose reduction and careful monitoring. Recommended dosing is variable: - Renal drug database recommends unstable angina and MI dose as in normal renal function. Other indications 1.5mg daily.¹³ If >100kg, it recommends initial dose of 10mg daily, then 7.5mg daily thereafter for treatment of a DVT.¹³ - Up to Date (ASHP Stacy et al 2018) recommends no adjustment, just close monitoring.¹⁴ - Warkentin 2011¹⁵ recommends normal dosing on day 1 then reduce dose depending on fondaparinux-specific anti-Xa levels. Often maintenance dose of only 2.5 to 5mg subcutaneously daily is required.
<30 mL/min	Total clearance is reduced ~55% compared to patients with CrCl >80 mL/minute. Limited data, relatively contraindicated, avoid.
Renal Replacement therapy	Limited data available (case reports and small case series). Dosing recommendations are largely based on expert opinion as practice is variable. ^{3,16} A consensus statement from the THANZ HIT Writing Group ³ recommends the following: 2.5 mg SC or 0.03–0.05 mg/kg IV before dialysis; monitor for drug accumulation.

Appendix 5: Bivalirudin^{3,17,18}

1.0 Availability

- Vial contains 250 mg of bivalirudin powder for injection.

2.0 Indications

- Treatment of HIT or use in patients with prior history of HIT are not current SMF listed indications. **Requires application for IPA.** Check the SMF for current approved indications. Use must be under the direction of Haematology.
- Exceptions: ACS requiring PCI with current or prior history of HIT.
 - Seek Cardiology advice and refer to the **G42 Bivalirudin guideline**.

3.0 Preparation

- Reconstitute each 250 mg vial with 5 mL of water for injection = 50 mg/mL solution.
- IV infusion: dilute 250 mg (5 mL) to 50 mL with a compatible fluid to make a concentration of 5 mg/mL.
- Refer to the AIDH for list of compatible and incompatible fluids and medicines, as well as Y-site compatibility.

4.0 Dose and Administration

- No bolus required.

Renal Function (CrCl mL/min)	Continuous Intravenous Infusion
CrCl > 60 mL/min	0.15 mg/kg/hr
CrCl 30-60 mL/min	0.08 mg/kg/hr

5.0 Renal Impairment <30 mL/min and Renal Replacement Therapy

- Relatively CI, Haematologist advice required.
- Refer to the Renal Drug Handbook and Kiser et. al. for more information¹⁷
- References, such as the Australian Medicines Handbook, provide dose recommendations for ACS requiring PCI with renal impairment CrCl < 30 mL/min and dialysis. The doses for PCI are higher than HIT.

6.0 Monitoring³

- Target APTT, 1.5–2.5 × patient baseline APTT.
- Measure APTT 4-hourly until two consecutive results in target range.
- Consider initial 2-hourly monitoring in patients at high risk of bleeding.
- The table below is a guide for dosing adjustment; this should be individualised¹⁷.

APTT (secs)	Dose Adjustment
<1.5 x baseline	Increase infusion by 20%
1.5-2.5 x baseline	Continue current rate
>2.5-3 x baseline	Decrease infusion by 20%
>3 x baseline	Hold for 1 hour, then restart at 50% lower rate

7.0 Cardiopulmonary bypass, on pump³

- Requires application for IPA.
- Before:** 1 mg/kg bolus, commence infusion at 2.5 mg/kg/h — target ACT 2.5 × baseline.
- Add 50 mg to pump circuit — avoid stasis in CPB circuit.
- During:** continue infusion at 2.5 mg/kg/h; monitor ACT every 30 min — target ACT 2.5 × baseline.
- Increase infusion rate only if ACT levels decrease below target or give repeated fractionated boluses of 0.25 mg/kg.

Appendix 6: Argatroban^{3,18}

1.0 Availability

- The Exembol® brand argatroban vial contains 250 mg/ 2.5 mL.
- Non TGA registered medicine – complete relevant Special Access Scheme forms.

THIS MEDICINE IS NOT ON FORMULARY, IS NON-TGA REGISTERED AND IS NOT STOCKED IN THE HOSPITAL. CONTACT THE CLINICAL WARD PHARMACIST FOR ADVICE IF YOU WISH TO PRESCRIBE.

2.0 Indications

- **ALL indications require application for IPA.**
- Treatment of HIT or use in patients with prior history of HIT.
- Option in severe renal impairment.
- Use must be under the direction of Haematology.

3.0 Preparation

- Dilute with a compatible fluid to produce a final concentration of approximately 1mg/mL (Note: doses are in microgram/kg/min, not mg), for example dilute 50 mg (0.5 mL) to 50 mL, 250 mg (2.5 mL) to 250 mL or 500 mg (5 mL) to 500 mL.
- Invert the bag several times to mix well.
- Use an infusion pump.
- Compatible fluids: Glucose 5%, Hartmann's, sodium chloride 0.9%.
- Refer to AIDH18 for more compatible and incompatible medicines, fluids as well as Y-site compatibility.

4.0 Dose for treatment of HIT and Administration

- No bolus required.

Clinical Situation		Continuous Intravenous Infusion
Non-critically unwell patients	Normal hepatic function	2 microg/kg/min
	Hepatic impairment	0.5 microg/kg/min
Critically unwell patients	Life threatening thromboembolism	2 microg/kg/min
	Non-life-threatening thromboembolism	1 microg/kg/min
	No thrombosis	0.5 microg/kg/min

5.0 Renal Impairment <50 mL/min and Renal Replacement Therapy

Continuous haemodialysis (3)

- 100 microg/kg bolus: initial infusion 0.5 microg/kg/min — target APTT 1.5–3.0 × baseline.

Intermittent haemodialysis (3)

- 250 microg /kg bolus: initial infusion 2.0 microg /kg/min — target APTT 1.5–3.0 × baseline or ACT 170–230s.

6.0 Hepatic Impairment

- Reduce dose in moderate to severe hepatic impairment (Child-Pugh class B and C).

7.0 Monitoring

- Target APTT, 1.5–3.0 × patient baseline APTT.
- Measure APTT 2-hourly until two consecutive results in target range and then a minimum of daily.
- Do not exceed APTT of 100s.